

SCC.160 – Fundamentals of Communication Systems

Revision Questions (Satellite Navigation)

KNOWLEDGE

1. What are the 3 **segments** of GPS called?
2. What is the **average life expectancy** of a GPS satellite? **How many** satellites comprise the GPS constellation?
3. What is the **function** of the **monitoring stations** and the **master control station** of the GPS network?
4. **How many satellites** (in theory) should be in view in order for a GPS receiver to provide satisfactory location information for **civilian** use? Would your answer be different if you considered GPS receivers for **aviation** use?
5. Explain how knowledge of local atmospheric conditions can improve GPS accuracy (**assisted GPS**).

6. Name **5 sources** of potential **errors**, which affect the **accuracy** of the GPS measurement.

UNDERSTANDING

1. Movies/TV often show people being followed using a “GPS tracker” or some similar device.
 - a. Can this be done? [**Yes / No**] (*be prepared to support your answer*)
 - b. Could you be traced when you use your in-car sat-nav device? [**Yes / No**]
 - c. Could you be traced when you use a mobile phone?
[**Yes / Only if a sat-nav application has been installed / No**]